

James N Leathers IV  
Everett, WA  
jleathers4@outlook.com

## Education

**Purdue University** - West Lafayette, IN

Major: **Electrical Engineering Technology**  
Minor: **Organizational Leadership**

Graduation: May 2026

## Work Experience

|                                                                                                                                                                                                                                                                    |                                        |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-------------|
| <b>Electronics Hardware Engineer &amp; PCB Design Intern</b>                                                                                                                                                                                                       | Hughes Network Systems, LLC & EchoStar | Summer 2025 |
| <ul style="list-style-type: none"><li>Reverse engineering legacy embedded system hardware for modernization integration with Cadence Allegro, Blueprint, &amp; Viewmate Software; presented completed technical documentation to company SVPs &amp; VPs.</li></ul> |                                        |             |
| <b>RF Applications Undergrad Lab Teaching Assistant</b>                                                                                                                                                                                                            | Wireless Communications                | 2025 - 2026 |
| <ul style="list-style-type: none"><li>Instructed classes on how to build &amp; conduct analysis of a superheterodyne receiver &amp; SMT components.</li></ul>                                                                                                      |                                        |             |
| <b>Software Undergrad Lab Teaching Assistant</b>                                                                                                                                                                                                                   | Intro to Digital Systems               | 2024 - 2025 |
| <ul style="list-style-type: none"><li>Instructed classes utilizing Microchip Studio C programming for weekly ATMEGA 2560-based projects.</li></ul>                                                                                                                 |                                        |             |
| Mentor - High School Programming Project                                                                                                                                                                                                                           |                                        | Spring 2024 |
| <ul style="list-style-type: none"><li>Mentored a 9th-grade high school student in C programming for an ATMEGA 328p-based science project.</li></ul>                                                                                                                |                                        |             |
| Handyman & Landscaping Assistant                                                                                                                                                                                                                                   | J Handyman, LLC                        | 2018 - 2021 |
| <ul style="list-style-type: none"><li>Assisted in minor electrical &amp; plumbing repairs; operated machinery on commercial &amp; residential properties.</li></ul>                                                                                                |                                        |             |

## Platforms

|                                                                                                                                                           |                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Cadence Allegro</b>                                                                                                                                    | Hardware Engineering/PCB Design Internship   |
| <ul style="list-style-type: none"><li>Designed embedded system hardware at Hughes Network Systems, LLC under NDA.</li></ul>                               |                                              |
| <b>Docker</b>                                                                                                                                             | Personal Projects                            |
| <ul style="list-style-type: none"><li>Deployed AI Surveillance &amp; Smart Automation containers for a real-time local network server.</li></ul>          |                                              |
| <b>Node-RED</b>                                                                                                                                           | Computer-Based Data Acquisition Applications |
| <ul style="list-style-type: none"><li>Developed data acquisition systems with Raspberry Pi 4 Linux OS, JavaScript, &amp; Python.</li></ul>                |                                              |
| <b>Microchip Studios &amp; STM32CubeIDE</b>                                                                                                               | Advanced Embedded Digital Systems            |
| <ul style="list-style-type: none"><li>Designed ARM Cortex-M4 based systems with STM32L476RG &amp; used ATMEGA 2560 &amp; 328P Microcontrollers.</li></ul> |                                              |

## Software

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|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>MATLAB Simulink</li></ul>         | (Adv. Digital Signal Processing – Designing DSP Filters & SDRs)                   |
| <ul style="list-style-type: none"><li>NI Multisim</li></ul>             | (DC & AC Power Electronics - simulation, prototypes, & calculations)              |
| <ul style="list-style-type: none"><li>SOLIDWORKS &amp; Fusion</li></ul> | (Adv. Embedded Digital Systems – Designed Components for 3D Print implementation) |
| <ul style="list-style-type: none"><li>KiCad &amp; EAGLE</li></ul>       | (Electronic Prototype Development & Adv. Embedded Digital Systems – Custom PCBs)  |
| <ul style="list-style-type: none"><li>Quartus II</li></ul>              | (Concurrent Digital Systems - FBGA board AHDL programming)                        |

## Projects

|                                                                                                                                                                                                                                                                                   |                                          |             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------|
| <b>AI Visual Defect Detection System</b>                                                                                                                                                                                                                                          | Cummins Inc. & Purdue Polytech Institute | 2025 - 2026 |
| <ul style="list-style-type: none"><li>Project Manager for industry-sponsored capstone; designed/investigated a non-contact inspection system for assembled manufactured parts, utilizing AI &amp; machine learning for improved quality control.</li></ul>                        |                                          |             |
| <b>AI Surveillance &amp; Smart Automation Server</b>                                                                                                                                                                                                                              |                                          |             |
| <ul style="list-style-type: none"><li>Developed Linux-based local network server utilizing Docker to host an AI-powered NVR for intelligent video surveillance with YAML; integrated Home Assistant, MQTT Broker, &amp; Zigbee comm. protocol for IoT stack automation.</li></ul> |                                          |             |
| <b>Spot Micro Quadruped Robot</b>                                                                                                                                                                                                                                                 |                                          |             |
| <ul style="list-style-type: none"><li>Designing &amp; currently building an open-sourced AI-piloted Spot Micro Robot Dog, inspired by Boston Dynamics' Spot; implementing computer vision, RF control, &amp; embedded control systems.</li></ul>                                  |                                          |             |
| <b>Signal Hacking</b>                                                                                                                                                                                                                                                             |                                          |             |
| <ul style="list-style-type: none"><li>Exploring NFC &amp; 125kHz RFID signal emulation with Flipper Zero.</li></ul>                                                                                                                                                               |                                          |             |
| Stereo Audio Amplifier +10W (University)                                                                                                                                                                                                                                          |                                          |             |
| <ul style="list-style-type: none"><li>Worked in a team environment to develop an embedded digital system for a jukebox project.</li></ul>                                                                                                                                         |                                          |             |
| Superheterodyne FM Radio Receiver (University)                                                                                                                                                                                                                                    |                                          |             |
| <ul style="list-style-type: none"><li>Developed &amp; tested subcircuits in wireless communications to build an FM radio with Class A amplifier.</li></ul>                                                                                                                        |                                          |             |
| Programmable Crane (University)                                                                                                                                                                                                                                                   |                                          |             |
| <ul style="list-style-type: none"><li>Worked in a team environment to develop EEPROM programmable crane.</li></ul>                                                                                                                                                                |                                          |             |

## Extracurriculars

|                                                                                                                                       |                |
|---------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Phi Kappa Tau Fraternity Member                                                                                                       | 2024 - Present |
| <ul style="list-style-type: none"><li>Executive Board Membership Orientation Officer (Implemented member education program)</li></ul> | 2024           |
| Tae Kwon Do – 2 <sup>nd</sup> Degree Black Belt Kukkiwon Dan Certificate (United States Martial Arts Academy)                         | 2009 - 2019    |

## Certifications

|                                                                                           |                |
|-------------------------------------------------------------------------------------------|----------------|
| CPR/AED For the Professional Rescuer Certificate (Purdue University & American Red Cross) | 2024 - Present |
| Jr. Open Water Diver Certificate (Professional Association of Diving Instructors   PADI)  | 2015 - Present |
| Library of Congress Researcher                                                            | 2024 - Present |

## Foreign Language

German - 3 years of study